Design and Implementation of a Line Impedance Stabilization Network (LISN)

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Chapter 1 Introduction

1.1 Document Objectives

A Line Impedance Stabilization Network (LISN) is an essential piece of equipment for evaluating the conducted electromagnetic emissions of electronic systems. It provides a measurement environment with a defined impedance while isolating emissions generated by the equipment under test from disturbances present on the power supply network. As a result, measurements become more repeatable and can be compared across different test sessions and laboratories.

The purpose of this document is to describe the complete design and implementation process of a LISN intended for laboratory research. The topics covered include the theoretical background, design requirements, component selection, hardware implementation, characterization, and procedures for using the equipment in conducted-emission measurements.

In addition to documenting the prototype that has been built, this document records the design decisions and their underlying calculations. This information is intended to support maintenance, further improvements, and the development of future LISN revisions.

1.2 Introduction to Electromagnetic Compatibility

The rapid development of modern electronic systems has led to a significant increase in potential sources of electromagnetic interference. Switching power supplies, inverters, high-speed microcontrollers, motor-drive circuits, wireless communication equipment, and power semiconductor devices can all generate electromagnetic disturbances during operation.

If left uncontrolled, these disturbances may propagate to other equipment through conductive connections or radiate into the surrounding space. They can degrade performance, corrupt data, destabilize systems, or even damage sensitive electronic equipment.

To ensure that electronic devices can operate simultaneously in the same environment without adversely affecting one another, the concept of **Electromagnetic Compatibility (EMC)** was introduced. EMC is the ability of a device to operate as intended in its electromagnetic environment without producing electromagnetic disturbances above the levels permitted for other equipment.

EMC covers two main aspects:

- **Electromagnetic Interference (EMI):** electromagnetic disturbances generated and

emitted by equipment.

- **Electromagnetic Susceptibility (EMS):** the ability of equipment to continue operating correctly when exposed to external electromagnetic disturbances.

A device can be considered EMC-compliant only when it satisfies both conditions: its emissions remain within the applicable limits, and it has sufficient immunity to disturbances from its environment.

During electronic product development, EMC evaluation is commonly carried out from the research and design stages. Early testing helps identify electromagnetic-emission problems and reduces the cost of modifications before the product enters formal certification testing.

1.3 Conducted Electromagnetic Emissions

Electromagnetic disturbances generated by electronic equipment can propagate through several mechanisms. Based on the propagation medium, electromagnetic emissions are generally divided into two main categories: conducted emissions and radiated emissions.

Conducted emissions are disturbances that propagate through conductive connections, such as cables linking equipment to a power source or connecting different devices. Since power cables can also act as unintended signal-transmission paths, these disturbances may travel over considerable distances and affect other equipment connected to the same electrical system.

In many EMC standards for commercial and industrial equipment, conducted-emission measurements are performed over the frequency range from $150\,\mathrm{kHz}$ to $30\,\mathrm{MHz}$. In this range, conductive paths can have a significant influence on electromagnetic-disturbance propagation. It is also the primary operating range considered for the LISN described in this document.

Conducted-emission evaluation is commonly performed during product development to identify dominant noise sources, optimize EMI filters, and verify the effectiveness of noise-mitigation measures before formal EMC certification testing.

1.4 Role of a LISN in Conducted-Emission Measurements

In a practical installation, the impedance of the power network is not constant. It depends on several factors, including cable length, power-distribution topology, connected loads, and source characteristics. The power network may also contain disturbances generated by other electrical equipment connected to the same system.

If the equipment under test is connected directly to the power source during an emission measurement, the result can depend strongly on the electrical conditions at the time of testing. This reduces measurement repeatability and makes it difficult to compare results obtained at different times or in different laboratories.

A Line Impedance Stabilization Network is designed to address this problem. A LISN establishes a defined impedance between the power source and the equipment under test while

isolating the measurement instrument from disturbances present on the supply network. It also provides a standardized measurement port through which high-frequency noise can be coupled to a spectrum analyzer or EMI receiver.

By standardizing the impedance presented by the supply network, a LISN improves the repeatability of conducted-emission measurements and supports EMC evaluation procedures used in both research and industry.

The operating principles and detailed structure of a LISN are presented in Chapter 2.

1.5 Scope of This Document

This document focuses on the design and implementation of a laboratory LISN intended for evaluating conducted emissions from low-power electronic power systems, switching power supplies, and embedded circuits.

The document covers the theoretical background, design methodology, hardware implementation, characterization, and operating instructions. Formal EMC certification procedures performed by accredited test laboratories are introduced only where necessary and are not the main focus of this work.

Chapter 2 Theoretical Background

2.1 Conducted Electromagnetic Emissions

In modern electronic systems, particularly switching-mode power supplies (SMPSs), inverters, motor controllers, and high-speed digital circuits, electromagnetic interference is virtually unavoidable. Power semiconductor devices such as MOSFETs, IGBTs, and GaN transistors operate through high-speed switching. Their switching edges exhibit high rates of change of voltage (dv/dt) and current (di/dt), making switching transitions one of the main sources of electromagnetic disturbances in power-electronic systems.

According to Fourier analysis, a switching waveform contains not only a component at the fundamental switching frequency but also numerous harmonic components at higher frequencies. Although a power converter may operate at only a few tens or hundreds of kilohertz, its noise spectrum can extend to tens or even hundreds of megahertz. Therefore, evaluating the spectral characteristics of these disturbances is an important part of system design and optimization.

Electromagnetic disturbances can propagate through several mechanisms. Based on the propagation medium, they are generally divided into two main categories:

- **Conducted Emission**: disturbances that propagate through conductive paths, such as power cables, signal cables, or grounding conductors.
- **Radiated Emission**: disturbances emitted as electromagnetic waves and propagated through space.

This document focuses on conducted emissions because they are the primary phenomenon that the Line Impedance Stabilization Network (LISN) is designed to evaluate.

For conducted emissions, power cables serve not only as energy-delivery paths but also as propagation paths for high-frequency disturbances. These components may travel from the Device Under Test (DUT) back toward the power source or propagate to other equipment connected to the same electrical system.

According to EMC standards such as CISPR 16 and CISPR 25 [2, 3], conducted-emission measurements are commonly performed over the frequency range from $150\,\mathrm{kHz}$ to $30\,\mathrm{MHz}$. This is a frequency range in which conducted disturbances are particularly relevant and is also the primary operating range considered for the LISN described in this document.

Accurately determining the conducted-emission level allows designers to evaluate EMI-filter

performance, identify dominant emission sources, and optimize hardware before proceeding to formal EMC certification testing.

2.2 Conducted-Noise Propagation Modes

Differential-Mode and Common-Mode Noise

Characteristic	Differential-Mode (DM) Noise	Common-Mode (CM) Noise
Definition	Noise that appears **in opposite phase** on two signal conductors (or between a signal conductor and its local return/ground conductor).	Noise that appears **in phase**, with approximately equal amplitude, on both conductors relative to a common ground reference.
Current path	Noise current flows along one conductor and **returns** through the other conductor, forming a closed loop.	Noise current flows along both signal conductors and **returns through the ground (GND) path** or parasitic earth paths.
Effect on a circuit	Directly affects the voltage difference between the two receiver inputs ($V_{in+} - V_{in-}$), thereby directly distorting the useful signal.	Affects both inputs simultaneously and is theoretically rejected by circuits with a sufficiently high common-mode rejection ratio (CMRR).

2.3 Challenges in Conducted-Emission Measurements

The objective of a conducted-emission measurement is to determine the high-frequency noise generated by the DUT and propagated back toward the power source. In practice, however, this measurement is not straightforward because the power-supply network itself has electrical characteristics that can vary over time and with the installation.

The impedance of the power network depends on several factors, including cable length, network topology, connected loads, connector contact quality, and source characteristics. Consequently, the same device may produce somewhat different measurement results when tested at different locations or at different times.

In addition, a power network contains disturbances from many sources, such as switching power supplies, electric motors, inverters, and other electronic equipment connected to the same system. These disturbances may overlap with the signal of interest, making it difficult to identify the emissions originating specifically from the DUT.

Another issue is that spectrum analyzers and EMI receivers are generally designed with a standard input impedance of $50\,\Omega$. Connecting such an instrument directly to a power line would expose it to potentially hazardous voltages and could also alter the electrical characteristics of the system, resulting in measurements that do not represent the DUT's normal operating conditions.

For these reasons, conducted-emission measurements require an intermediate network that isolates the measurement instrument from the power source while establishing a defined and stable impedance throughout the test. This function is provided by the Line Impedance Stabilization Network (LISN), which is discussed in detail in the following sections.

2.4 Introduction to the Line Impedance Stabilization Network

As discussed in Section 2.3, the impedance of a practical power-supply network is not constant. It varies with the network topology, cable length, source characteristics, and loads operating on the same system. Therefore, if the Device Under Test (DUT) is connected directly to the power network for conducted-emission measurements, the results may depend significantly on the supply conditions at the time of the test.

To eliminate this dependency, an impedance-stabilizing network is inserted between the power source and the DUT. This network is known as a **Line Impedance Stabilization Network (LISN)**.

A LISN is a measurement device used in conducted-emission testing to establish a defined and repeatable electrical environment. Instead of allowing the DUT to operate from a source with an uncontrolled impedance, the LISN presents a specified impedance over the measurement frequency range. This helps ensure that the measurement results are not unduly affected by variations in the power-supply network.

In addition to impedance stabilization, the LISN separates low-frequency power delivery from the high-frequency disturbances generated by the DUT. This allows the DUT to operate normally from an AC or DC supply while coupling RF noise components to a spectrum analyzer or EMI receiver for measurement.

The LISN can be regarded as an interface between three elements of the measurement system:

1. The power source supplying energy to the DUT.
2. The Device Under Test (DUT).
3. The noise-measurement instrument, typically a spectrum analyzer or EMI receiver.

During operation, a LISN must satisfy several technical requirements simultaneously. First, it must allow power-frequency current (at 50 Hz or 60 Hz) to reach the DUT with low voltage drop so that the DUT's operating conditions are not significantly affected. Second, high-frequency disturbances generated by the DUT must be directed toward the RF measurement port rather than propagating back to the power source. Finally, the measurement port must provide an impedance compatible with RF test instruments, which is typically standardized at $50\,\Omega$.

By satisfying these requirements, the LISN reduces the influence of the power network on conducted-emission measurements and enables tests with improved repeatability. For this reason, LISNs are standard components in many EMC evaluation systems, from laboratory research to formal testing against international standards.

Depending on the applicable test standard and application, LISN designs may differ in inductance, operating frequency range, rated current, and connection configuration. Nevertheless, their fundamental principle is the same: to stabilize the impedance of the power line and separate high-frequency disturbances from the power-delivery path.

2.5 Operating Principle of the Line Impedance Stabilization Network

Having established the role of a LISN in a conducted-emission measurement system, this section examines its operating principle. Although the detailed structure may vary with the applicable standard and the DUT power level, LISN configurations are based on the same fundamental concept: **separating the power-delivery path from the high-frequency noise path**.

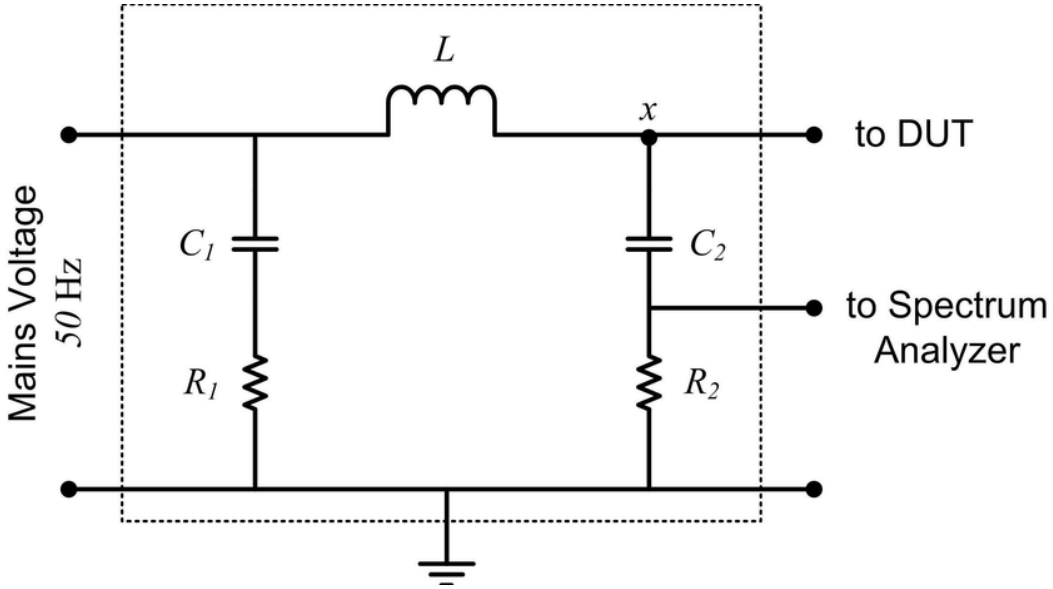


Figure 2.1: Basic schematic of a Line Impedance Stabilization Network.

Figure 2.1 shows a general LISN schematic. The device is placed between the power source and the DUT and provides an RF port for connection to a spectrum analyzer or EMI receiver. Although a practical circuit may include additional protection and coupling components, its overall operating principle is primarily determined by the frequency-dependent characteristics of the inductor and capacitor.

First, consider a power source operating at the mains frequency of $50\,\mathrm{Hz}$. At this frequency, the inductor has a very low impedance compared with the load impedance. Consequently, power is delivered to the DUT with only a small voltage drop across the LISN. This ensures that inserting the LISN into the measurement setup does not significantly alter the DUT's operating conditions.

In contrast, disturbances generated by the DUT typically occupy a frequency range from hundreds of kilohertz to several tens of megahertz. As frequency increases, the inductor impedance increases according to

$$X_L = 2\pi f L \tag{2.1}$$

Meanwhile, the impedance of a capacitor is given by

$$X_C = \frac{1}{2\pi f C} \tag{2.2}$$

Thus, as frequency increases, the inductor increasingly impedes noise current flowing back toward the source, whereas the capacitor provides a lower-impedance path for high-frequency components.

Through the combination of these two components, power-frequency current and high-frequency noise current are directed along different paths. Power-frequency current continues to flow toward the DUT, while high-frequency noise is coupled to the RF measurement port instead of propagating back into the power network.

To ensure compatibility with RF test instruments, the LISN measurement port is standardized at $50\,\Omega$. This impedance is widely used by spectrum analyzers, EMI receivers, and vector network analyzers. Maintaining the standard impedance reduces signal reflections and helps make measurements comparable across different instruments and laboratories.

A LISN is therefore not simply a conventional EMI filter. An EMI filter is intended to suppress disturbances propagating into the surrounding system, whereas a LISN is intended to establish a defined measurement impedance and provide a standardized path for acquiring noise generated by the DUT. For this reason, the measured noise level may, in some cases, be higher with a LISN than when the DUT is connected directly to the power source. The LISN measurement, however, represents the DUT's emissions under the defined measurement conditions.

Most LISNs, despite variations in test standards and DUT power ratings, consist of several basic functional blocks. Each component has a specific role in stabilizing the power-line impedance, isolating disturbances from the supply network, or coupling the noise signal to the measurement instrument.

The following subsections discuss the role of each component based on the schematic presented in Section 2.5.

2.5.1 Inductor

The inductor is one of the most important components in a LISN. Its primary function is to create an impedance difference between the power-supply frequency and the frequency range of the noise signal.

At the power-line frequency of $50\,\mathrm{Hz}$, the inductor's reactance is very small. Therefore, nearly all of the power supplied by the source reaches the DUT without significantly changing its operating conditions.

At frequencies in the hundreds of kilohertz or tens of megahertz range, however, the inductor's reactance increases linearly with frequency. This limits the propagation of noise toward the source and directs high-frequency energy along the path provided within the LISN.

$$X_L = 2\pi f L \tag{2.3}$$

In a LISN design, the inductance is not selected arbitrarily; it is determined by the relevant EMC test standard. This value determines the impedance characteristics of the network over the measurement frequency range and directly affects the accuracy of conducted-emission measurements.

2.5.2 Coupling Capacitor

The coupling capacitor transfers high-frequency noise from the power line to the RF measurement port while blocking the supply voltage from reaching the measurement instrument.

The impedance characteristic of a capacitor is opposite to that of an inductor. Its reactance decreases as frequency increases, according to

$$X_C = \frac{1}{2\pi f C} \tag{2.4}$$

At the power-supply frequency, the capacitor's reactance is very high, so almost no power-frequency current flows through the measurement branch. Within the conducted-emission measurement band, however, its impedance decreases significantly, providing a path for noise signals to reach the spectrum analyzer.

The capacitance must be selected by balancing high-frequency signal coupling against the requirement to isolate the supply voltage from the measurement instrument. LISN designs commonly use safety-rated capacitors or capacitors with an appropriately high voltage rating to ensure safe operation.

2.5.3 $50\,\Omega$ Termination Network

RF test instruments such as spectrum analyzers, EMI receivers, and vector network analyzers are generally designed with a standard input impedance of $50\,\Omega$. The LISN measurement port must therefore maintain the same impedance to ensure compatibility with the instruments in the measurement system.

If the impedance at the measurement port is not properly controlled, the noise signal may be reflected at the connection point. This can distort the measured spectral amplitude and reduce measurement repeatability.

In addition to impedance matching, the resistor network contributes to establishing the standardized impedance presented to the DUT over the measurement frequency range. This is one of the key factors that allows conducted-emission results to be compared across different laboratories.

2.6 LISN Configurations

Although all LISNs are intended to stabilize the source impedance and provide a measurement port for conducted disturbances, several configurations are available depending on the equipment under test and the applicable EMC standard. These configurations differ in circuit topology, inductance, and the number of supported power lines.

2.6.1 Classification by Topology

Based on circuit topology, LISNs are commonly divided into three main types:

- **V-LISN (Voltage LISN):** This is one of the most widely used configurations for conducted-emission measurements. A V-LISN measures the noise voltage between each power conductor and ground. The measured voltage may contain both common-mode and differential-mode components.
- **Delta LISN:** A Delta LISN provides separate outputs for common-mode and differential-mode noise. In practice, a similar function can also be implemented by combining two V-LISNs with a signal-separation network (LISN Mate).
- **T-ISN (Telecommunication ISN):** Unlike the two configurations above, a T-ISN is designed for data lines or telecommunication cables rather than power-supply lines. It is primarily used to measure common-mode disturbances on twisted pairs.

2.6.2 Classification by Inductance

Another classification is based on the inductance of the impedance-stabilization network. Two commonly used configurations are the $5\,\mu\mathrm{H}$ LISN and the $50\,\mu\mathrm{H}$ LISN.

Table 2.1: Comparison of two common LISN configurations

Characteristic	$5\,\mu\mathrm{H}$ **LISN**	$50\,\mu\mathrm{H}$ **LISN**
Application	Automotive electronic equipment	Residential and industrial equipment
Typical representation	Short power cable (approximately 5 m)	Power wiring in building electrical systems
Typical frequency range	150 kHz–110 MHz	9 kHz–30 MHz
Commonly associated standard	CISPR 25	CISPR 16

According to Tekbox documentation, $5\,\mu\mathrm{H}$ LISNs are primarily used for EMC testing of automotive electronic equipment, whereas $50\,\mu\mathrm{H}$ LISNs are widely used for equipment operating

in residential and industrial environments. In addition to their different inductance values, the two configurations have different applicable frequency ranges and standard requirements.

2.6.3 Classification by Number of Power Lines

LISNs can also be classified according to the number of power lines incorporated into a single unit.

- **Single-path LISN**: used for a single DC power conductor or an individual line under test.
- **Two-path LISN**: commonly used for DC systems with positive and negative supply conductors, or single-phase AC systems with line and neutral conductors.
- **Four-path LISN**: used in three-phase power systems with three phase conductors and one neutral conductor.

The LISN developed in this document is a single-path V-LISN with a nominal inductance of $5\,\mu\mathrm{H}$. This configuration is suitable for conducted-emission investigations on DC-powered electronic systems and is intended to support measurements associated with CISPR 25.

Chapter 3 LISN Design

Following the discussion of the fundamental principles of conducted-emission measurements and the structure of the Line Impedance Stabilization Network (LISN) in Chapter 2, this chapter focuses on the hardware design of the LISN developed in this project.

The chapter discusses the design requirements, component-selection criteria, and factors that affect LISN performance over the measurement frequency range. Based on these considerations, each functional block is described, including the impedance-stabilization network, RF coupling circuit, protection circuit, and PCB design requirements.

3.1 Design Requirements

Designing a Line Impedance Stabilization Network (LISN) involves more than simply selecting component values from a schematic. The device must also satisfy the electrical, mechanical, and measurement requirements associated with its intended application. These requirements form the basis for selecting the circuit configuration, components, and hardware implementation approach discussed in the following sections.

In this project, the LISN is developed for laboratory research and conducted-emission evaluation of DC-powered electronic systems. The design aims to be practical to build, reasonably priced, and easy to adjust, while retaining the essential characteristics required for pre-compliance measurements.

The main design requirements are defined as follows:

- Use a V-LISN configuration with a nominal inductance of $5\,\mu\mathrm{H}$, suitable for conducted-emission measurements on DC-powered electronic systems.
- Operate over the frequency range from $150\,\mathrm{kHz}$ to $110\,\mathrm{MHz}$, supporting CISPR 25 pre-compliance investigations.
- Maintain a standard impedance of $50\,\Omega$ at the RF port to allow direct connection to a spectrum analyzer or EMI receiver.
- Supply current to the DUT without causing a significant voltage drop in the power path or disturbing the DUT's normal operating conditions.
- Minimize the influence of external power-source noise on the measurement results while coupling noise generated by the DUT to the measurement port with low attenuation.

- Provide electrical protection through components such as fuses, varistors, and discharge resistors to reduce risks to users and measurement equipment.
- Prefer commercially available components that are easy to source, fabricate, and maintain in a laboratory environment.
- Minimize PCB parasitics, including parasitic inductance and capacitance, while maintaining suitable insulation clearances for the system operating voltage.

Table 3.1: Design requirements of the LISN

Parameter	**Value**	**Purpose**
Nominal inductance	5 μH	Stabilize the source impedance according to the selected configuration
Operating frequency range	150 kHz–110 MHz	Support conducted-emission measurements
RF-port impedance	50 Ω	Ensure compatibility with RF measurement equipment
RF connector type	BNC or SMA	Connect to a spectrum analyzer
Supply type	DC	Match the intended equipment under test
Protection level	Limiter + Filter + Attenuator	Improve operational safety

3.2 Design of the Impedance-Stabilization Network

The impedance-stabilization network is the most important functional block of a Line Impedance Stabilization Network (LISN). It determines the impedance presented to the Device Under Test (DUT) during measurement and isolates high-frequency disturbances from the power source so that the noise signal can be acquired at the RF port.

As discussed in Chapter 2, the LISN impedance must remain stable over the measurement frequency range to ensure repeatability and consistency in conducted-emission measurements. Therefore, selecting the inductance value and the topology of the impedance-stabilization network are among the most important design decisions.

In this project, the LISN is based on the $5\,\mu\mathrm{H}$ configuration. This configuration is commonly used for conducted-emission measurements on DC-powered electronic systems and is consistent with the intended application of the prototype.

The impedance-stabilization network consists of an inductor connected in series with the power-supply line and supporting components that establish the impedance characteristics of the complete network. The inductor is the primary component responsible for differentiating between low-frequency power signals and high-frequency noise components.

At the supply frequency, the inductor reactance is very small, so the voltage supplied to the DUT is barely affected. As the frequency increases into the conducted-emission measurement band, the reactance rises and limits the propagation of noise toward the source. As a result, a larger portion of the high-frequency noise energy is directed toward the RF coupling network for measurement.

In addition to nominal inductance, the performance of the impedance-stabilization network is affected by practical component characteristics, including DC resistance (DCR), self-resonant frequency (SRF), saturation current, and parasitic parameters. These factors must be considered together when selecting components to maintain the intended LISN impedance over the operating frequency range.

The following sections describe the selection of the inductance value, the inductor-selection criteria, and practical considerations for hardware implementation.

3.2.1 Selection of the Inductance Value

For conducted-emission measurements on AC-powered electronic equipment, the $50\,\mu\mathrm{H}$ LISN configuration is commonly used to represent the impedance of residential or industrial power-distribution systems. In contrast, for DC-powered electronic systems, particularly automotive applications, CISPR 25 specifies the use of a $5\,\mu\mathrm{H}$ LISN configuration. The lower inductance value represents the relatively short power wiring and low source impedance found in vehicle electrical systems.

In this project, the LISN is developed for laboratory research and conducted-emission evaluation of DC-powered electronic systems. Therefore, the $5\,\mu\mathrm{H}$ configuration is selected as the design basis to align with the CISPR 25 measurement approach and the intended application of the device.

Table 3.2: Rationale for selecting the $5\,\mu\mathrm{H}$ LISN configuration

Criterion	**Reason for selection**
Equipment under test	DC-powered electronic equipment
Reference standard	CISPR 25
Nominal inductance	$5\,\mu\mathrm{H}$
Operating frequency range	$150\,\mathrm{kHz}$–$110\,\mathrm{MHz}$
Objective	Conducted-emission measurements under laboratory conditions

3.2.2 Inductor Selection

After defining the nominal inductance as $5\,\mu\mathrm{H}$, the next step is to select an implementation approach. Two common options are considered: using commercially available inductors or manufacturing an air-core inductor.

Air-core inductors are often used in EMC measurement equipment because of their linear

characteristics, absence of magnetic saturation, and negligible core losses. In addition, designers can directly adjust the number of turns, coil diameter, and spacing between turns to optimize the inductance and self-resonant frequency (SRF).

However, this approach requires suitably sized wire, fabrication tools, and measurement equipment to adjust the inductance after winding. For the initial prototype in this project, manufacturing a custom inductor would significantly increase implementation time and equipment requirements.

Commercial power inductors, on the other hand, are available in a range of $5\,\mu\mathrm{H}$ options, with high current ratings and frequency characteristics suitable for conducted-emission measurements. Using an off-the-shelf component shortens fabrication time, improves consistency between revisions, and simplifies replacement when necessary.

Table 3.3 compares the two approaches.

Table 3.3: Comparison of the two inductor implementation approaches

Criterion	**Commercial Inductor**	**Custom-Wound Inductor**
Implementation time	Short	Longer
Repeatability	High	Depends on the fabrication process
Parameter optimization	Limited	Highly flexible
Equipment requirements	Minimal	Requires tools and measurement equipment
Replaceability	Easy	Requires a new fabrication process
Suitability for an MVP	High	Moderate

Commercial Inductor Approach

A commercial inductor is selected to reduce design time, simplify fabrication, and ensure repeatability between test iterations. Commercial power inductors are manufactured using industrial processes and are supplied with detailed specifications, including current-handling capability and frequency characteristics. These properties make them suitable for a laboratory LISN prototype.

In this design, a Würth Elektronik WE-HCC series power inductor, part number 7443340100, is selected as the reference component. It uses a ferrite core, a wirewound construction, and a surface-mount device (SMD) package. Five inductors are connected in series to obtain a total nominal inductance of $5\,\mu\mathrm{H}$.

The main component specifications are listed in Table 3.4.

Table 3.4: Specifications of the selected commercial inductor

Parameter	Value
Nominal inductance	$1\,\mu\mathrm{H}$
Tolerance	$\pm 20\%$
Continuous operating current	$17\,\mathrm{A}$
Saturation current	$24\,\mathrm{A}$
DC resistance (DCR)	$3.9\,\mathrm{m}\Omega$ (Max.)
Self-resonant frequency (SRF)	$147\,\mathrm{MHz}$
Core material	Ferrite
Package type	SMD

Among these specifications, the self-resonant frequency (SRF) is one of the most important criteria when selecting an inductor for a LISN. With an SRF of approximately $147\,\mathrm{MHz}$, above the upper limit of the measurement range considered here ($110\,\mathrm{MHz}$), the inductor is expected to retain inductive behavior throughout the intended operating band. This helps limit the influence of parasitic capacitance and maintain the impedance characteristics of the network.

In addition, a low DC resistance of approximately $3.9\,\mathrm{m}\Omega$ reduces the voltage drop in the power path and limits conduction losses as the load current increases. The continuous current rating of up to $17\,\mathrm{A}$ also provides adequate margin for the system during research and testing.

Although ferrite-core inductors may experience magnetic saturation at high current or during overcurrent transients, the component's saturation-current rating of $24\,\mathrm{A}$ is significantly higher than the expected operating current of the system. Therefore, the effect of magnetic saturation is expected to be negligible under normal operating conditions.

Custom Air-Core Inductor Approach

In addition to using a commercial inductor, a second approach considered is the fabrication of a custom air-core inductor. This method is commonly used in RF and EMC equipment when the frequency response must be optimized or when the required inductance is not readily available as an off-the-shelf component.

Unlike ferrite-core inductors, air-core inductors do not exhibit magnetic saturation and do not incur losses in a magnetic core. Their inductance therefore remains stable over the operating-current range, while their frequency response is mainly determined by winding geometry and parasitic parameters.

In the proposed design, the inductor is made from round copper wire with a diameter of approximately $1.6\,\mathrm{mm}$. The wire is wound around a former manufactured using 3D printing. The former is intended to maintain consistent coil diameter, turn spacing, and coil length during fabrication.

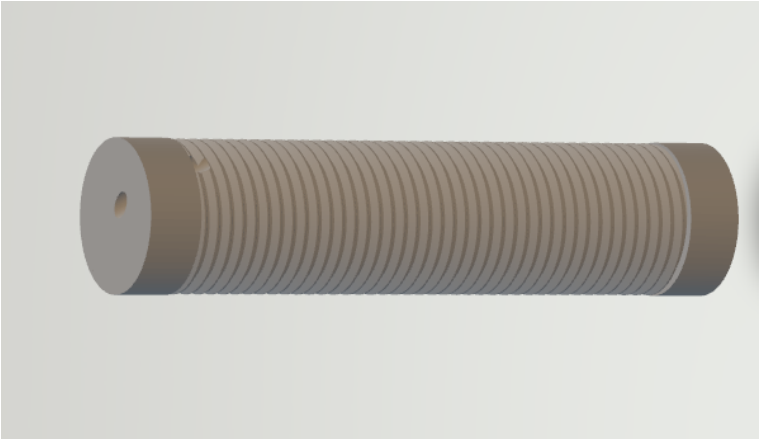


Figure 3.1: Proposed 3D-printed inductor former.

The coil inductance can be estimated using empirical formulas for air-core inductors. One commonly used expression is Wheeler's formula [4]:

$$L(\mu H) = \frac{r^2 N^2}{9r + 10l} \tag{3.1}$$

where:

- L is the coil inductance (μH);
- r is the mean coil radius (inches);
- l is the coil length (inches);
- N is the number of turns.

This formula provides an initial estimate of the required number of turns before fabrication. Once the coil has been manufactured, its actual inductance is measured using an LCR meter and adjusted, if necessary, by changing the number of turns or the spacing between turns.

An important advantage of this approach is that the designer can optimize the coil geometry to increase the self-resonant frequency (SRF) and reduce parasitic capacitance between turns. This is particularly relevant to conducted-emission measurement applications, where the inductor frequency response directly affects the impedance characteristics of the stabilization network.

However, custom fabrication also has several limitations. It requires suitable mechanical tools, measurement equipment to tune the inductance after winding, and careful manufacturing to achieve repeatability. In addition, an air-core inductor is generally much larger than a commercial component with the same nominal inductance.

3.3 Design and Simulation

Before hardware fabrication, the LISN design is evaluated using frequency-domain circuit simulations. The purpose of the simulation is to assess the electrical characteristics of the complete LISN network rather than relying only on the nominal specifications of individual components. In particular, inductance, capacitance, load resistance, and parasitic elements can all affect the frequency response of the circuit.

The simulations are performed in LTspice using an equivalent circuit model of a $5\,\mu\mathrm{H}$ LISN intended for DC-powered systems. The evaluation focuses on two main characteristics: the impedance seen from the DUT side and the reverse transmission from the DUT port toward the power-supply side.

The model is derived from the LISN schematic and includes the main components as well as selected parasitic elements required to approximate the behavior of the physical circuit. Additional blocks, such as the attenuator, limiter, and RF filter, are considered separately to avoid affecting the evaluation of the basic LISN characteristics.

3.3.1 Simulation Model

The simulations use an equivalent-circuit model of the $5\,\mu\mathrm{H}$ LISN. The model includes the main inductor, capacitors, damping resistors, a standard $50\,\Omega$ load at the RF port, and parasitic elements introduced to approximate the characteristics of the practical circuit.

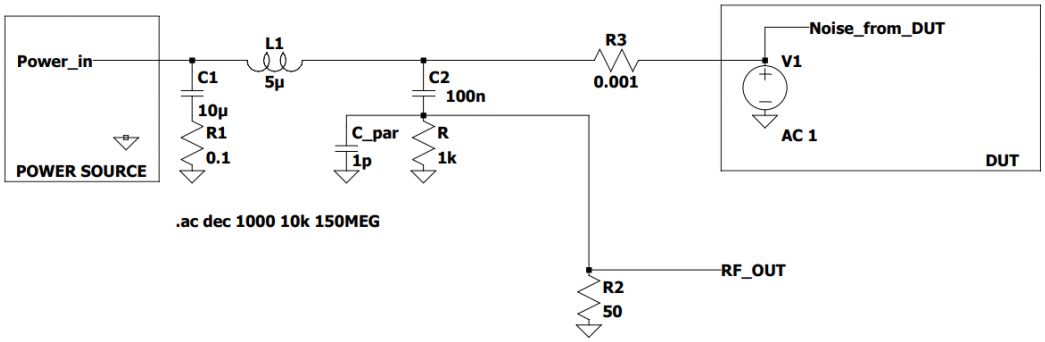


Figure 3.2: Equivalent LISN model used in the simulations.

In the model, an AC excitation source is connected to the DUT side and labeled `Noise_from_DUT`. Its AC amplitude is set to $1\,\mathrm{V}$ to evaluate the frequency response of the LISN. A $1\,\mathrm{m}\Omega$ resistor is connected in series between the excitation source and the LISN network to sense the current entering the circuit. This resistance is very small compared with the characteristic impedance of the system, so its influence on the overall response is negligible.

At the RF port, a standard $50\,\Omega$ load is used to represent the input impedance of the measurement instrument. Additional resistive and parasitic-capacitance branches are included at the RF port to investigate their influence on the high-frequency response.

The AC analysis is configured with a sweep from $10\,\mathrm{kHz}$ to $150\,\mathrm{MHz}$, using 1000 points

per decade. The primary frequency range considered in the evaluation is from $150\,\mathrm{kHz}$ to $110\,\mathrm{MHz}$.

Based on this model, different quantities are observed at selected circuit nodes while the circuit topology and excitation conditions are kept unchanged. This consistent setup allows the LISN characteristics to be evaluated using the same model and operating conditions.

3.3.2 Simulation of the Impedance Seen from the DUT Side

The impedance seen from the DUT side is one of the key LISN parameters. It represents the impedance presented to the equipment under test at the connection port. This characteristic directly affects conducted-emission measurements because the LISN is intended to provide a defined and stable impedance environment over the frequency range of interest.

To evaluate this parameter, an AC simulation is performed by applying an excitation source at the DUT port and measuring both the voltage at this port and the current flowing into the LISN network.

Simulation Setup

An AC voltage source with an amplitude of $1\,\mathrm{V}$ is connected to the node `Noise_from_DUT`. The current-sensing resistor R3, with a value of $1\,\mathrm{m\Omega}$, is connected in series between the source and the LISN. The current through R3 is the excitation current entering the network; therefore, `I(R3)` can be used to determine the equivalent impedance of the circuit.

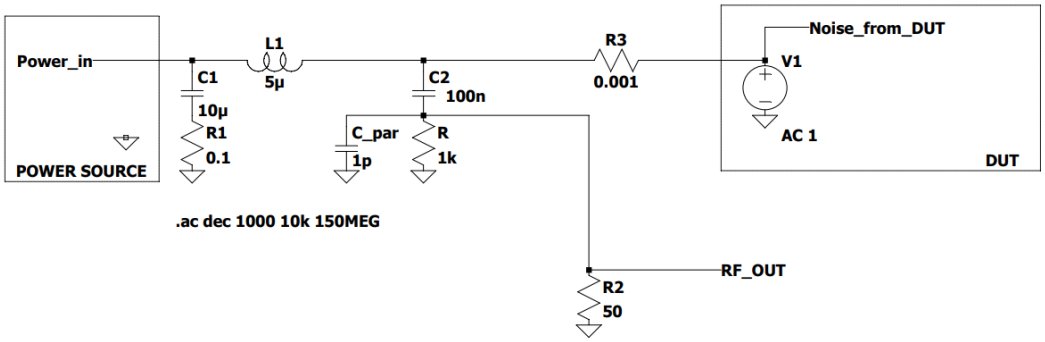


Figure 3.3: Excitation-source and current-measurement setup for determining the impedance seen from the DUT side.

According to Ohm's law, the equivalent impedance seen by the excitation source is determined by the ratio of voltage to current:

$$Z_{\mathrm{meas}}(f) = \frac{V_{\mathrm{DUT}}(f)}{I_{\mathrm{R3}}(f)} \tag{3.2}$$

where:

- $V_{\mathrm{DUT}}(f)$ is the voltage at node `Noise_from_DUT`;

- $I_{\mathrm{R3}}(f)$ is the current through the current-sensing resistor R3;
- $Z_{\mathrm{meas}}(f)$ is the equivalent impedance, including R3.

Since R3 is connected in series with the LISN network, the impedance of the LISN itself can be obtained from:

$$Z_{\mathrm{LISN}}(f) = \frac{V_{\mathrm{DUT}}(f)}{I_{\mathrm{R3}}(f)} - R_3 \tag{3.3}$$

where $R_3 = 1\,\mathrm{m\Omega}$. In this case, R3 is very small compared with the nominal impedance of $50\,\Omega$, so its effect on the result is negligible. When defining the expression in LTspice, the current reference direction through R3 must be set from the excitation source into the LISN network.

In addition to impedance magnitude, the phase angle is monitored to evaluate whether the network exhibits predominantly inductive or capacitive behavior as a function of frequency. The complex impedance can be expressed as:

$$Z_{\mathrm{LISN}}(f) = R(f) + jX(f) \tag{3.4}$$

where $R(f)$ is the resistive component and $X(f)$ is the reactance. The impedance phase angle is given by:

$$\varphi(f) = \arg\left(Z_{\mathrm{LISN}}(f)\right) \tag{3.5}$$

With the current reference direction defined above, a positive phase angle indicates predominantly inductive behavior, whereas a negative phase angle indicates predominantly capacitive behavior.

Result Evaluation

The simulation results are presented as impedance-magnitude and phase-angle plots versus frequency in Figure 3.4. These two quantities allow the impedance level and electrical behavior of the LISN network to be evaluated simultaneously over the frequency range of interest.

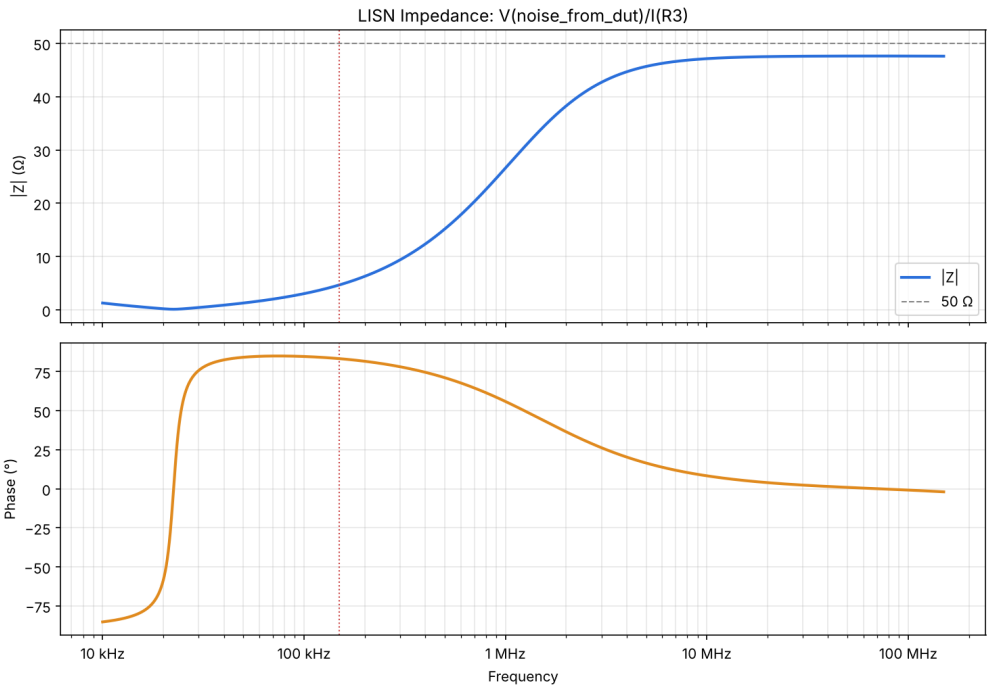


Figure 3.4: Magnitude and phase characteristics of the impedance seen from the DUT side, obtained from AC simulation.

Over the frequency range of interest, impedance variations reflect the combined effects of the inductor, capacitors, load resistors, and parasitic elements in the model. Observing both magnitude and phase makes it possible to identify frequency regions where the network characteristics change significantly, such as resonance regions or transitions between inductive and capacitive behavior.

The simulation results are used to evaluate the basic impedance characteristics of the LISN before additional functional circuits are integrated.

3.3.3 Simulation of Reverse Transmission from the DUT to the Power Source

In addition to impedance characteristics, the ability to limit high-frequency noise transmission from the DUT back toward the power source must also be investigated. A LISN should provide a path for noise from the DUT to reach the RF measurement port while limiting the propagation of these noise components toward the supply.

In this model, reverse transmission is evaluated by applying the excitation source on the DUT side and observing the voltage appearing at the power-supply input. This setup directly evaluates the voltage transferred from the DUT side to the source side under the selected simulation conditions.

Simulation Setup

The $1\,\mathrm{V}$ AC source remains connected to node `Noise_from_DUT`. The voltage at the power-supply input, labeled `Power_in`, is used as the output quantity to evaluate the signal transmitted back through the LISN.

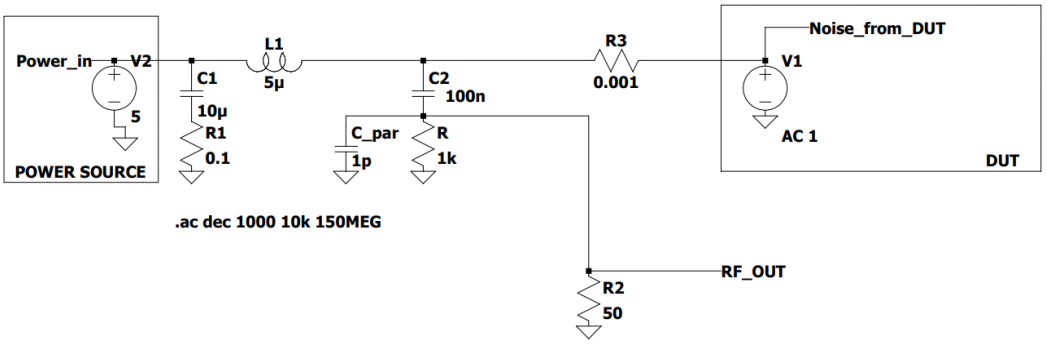


Figure 3.5: Simulation setup for evaluating reverse transmission from the DUT side toward the power-supply input.

The voltage transfer function from the DUT side toward the power source is defined as:

$$H_{\mathrm{rev}}(f) = \frac{V_{\mathrm{Power_in}}(f)}{V_{\mathrm{Noise_from_DUT}}(f)} \tag{3.6}$$

The reverse attenuation is expressed in decibels as:

$$IL_{\mathrm{rev}}(f) = -20\log_{10}\left|\frac{V_{\mathrm{Power_in}}(f)}{V_{\mathrm{Noise_from_DUT}}(f)}\right| \tag{3.7}$$

where:

- $V_{\mathrm{Noise_from_DUT}}(f)$ is the excitation voltage on the DUT side;
- $V_{\mathrm{Power_in}}(f)$ is the voltage measured at the power-supply input;
- $IL_{\mathrm{rev}}(f)$ is the reverse attenuation in dB.

A larger value of IL_{rev} indicates a smaller fraction of the voltage being transferred from the DUT side to the power source. This expression is used to generate the reverse-transmission characteristic versus frequency in LTspice.

Result Evaluation

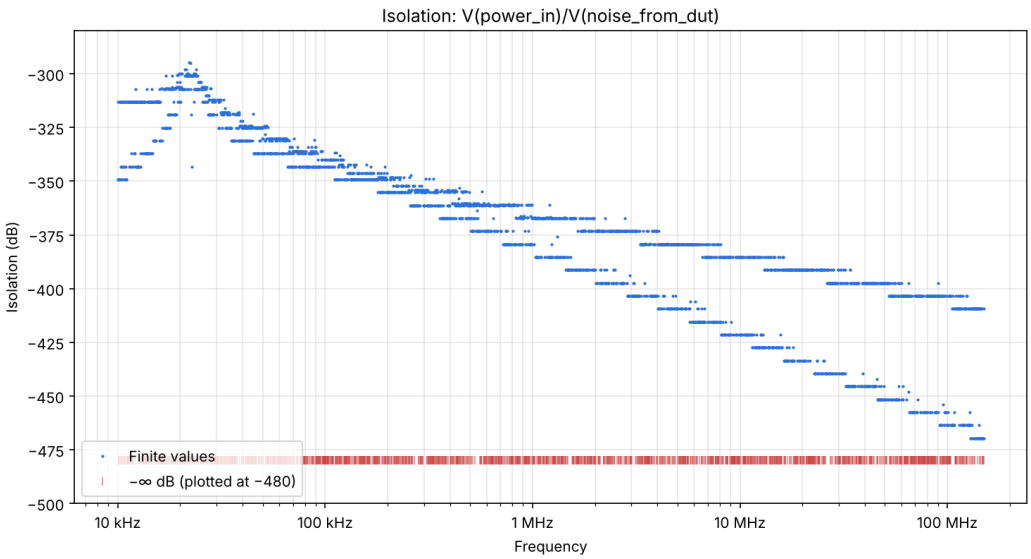


Figure 3.6: Simulated isolation characteristic between the power-supply input and the DUT-side port of the LISN.

The reverse-transmission curve indicates the level of the signal appearing at the power-supply input when the DUT-side port is excited, across the simulated frequency range. Its frequency dependence reflects the effects of the LC network and the loads in the LISN.

Frequency regions with lower attenuation require attention because a larger fraction of the DUT-side signal may be transmitted toward the source in those regions. Conversely, higher attenuation indicates better suppression of reverse transmission under the corresponding simulation conditions.

It should be noted that this result represents the reverse voltage-transfer characteristic of the simulated configuration. Its value depends on the excitation-source location, the loads, and the termination conditions at the ports. Therefore, when comparing it with a standards-based isolation test, the source and load configuration must be consistent with the conditions specified for that test.

3.3.4 Discussion

The AC simulations evaluate two important characteristics of the LISN network: the impedance seen from the DUT side and the ability to limit reverse signal transmission toward the power source. Applying the excitation source at the DUT port provides a consistent excitation condition when investigating different quantities. The impedance is determined from the voltage-to-current ratio, whereas reverse transmission is determined from the voltage ratio between the two ends of the network.

The results provide a basis for evaluating the basic LISN model before proceeding to hardware fabrication and verification. The effects of additional functional blocks, such as the attenuator, limiter, and filter, are considered separately to determine their influence on the overall system

characteristics.

3.4 Protection-Circuit Design

After the high-frequency noise has been separated from the power line by the LISN, the signal still requires conditioning before it is applied to the input of the spectrum analyzer. During conducted-emission measurements, the instrument input may be exposed to transient overvoltages or high-amplitude signals generated by the DUT. The measurement path must also provide AC coupling and maintain a stable transmission response over the intended operating frequency range.

To meet these requirements, an RF front-end circuit is placed between the LISN RF port and the spectrum analyzer, as shown below.

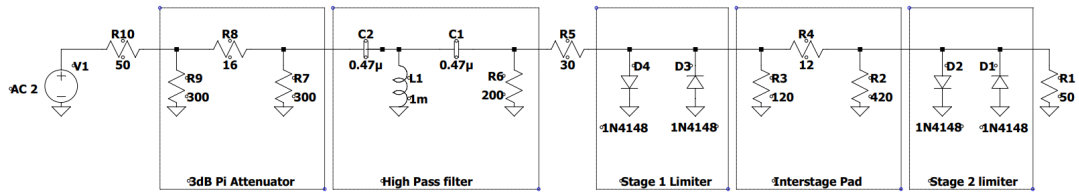


Figure 3.7: Functional blocks of the RF front-end circuit.

To evaluate the response of the complete circuit, an AC simulation is performed using a $2\,\mathrm{V}$ excitation source, a source impedance of $50\,\Omega$, and a standard $50\,\Omega$ load representing the input impedance of the spectrum analyzer. The frequency sweep covers the range from $100\,\mathrm{Hz}$ to $300\,\mathrm{MHz}$.

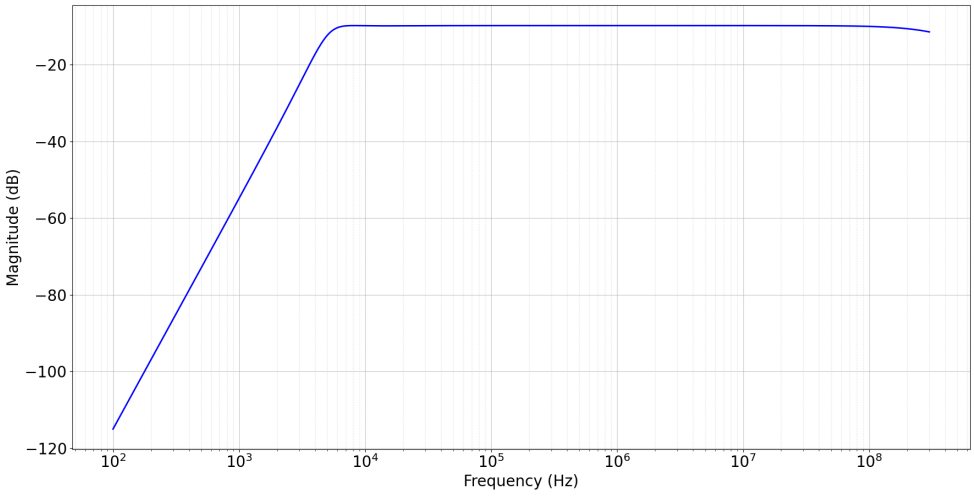


Figure 3.8: Transmission response of the protection circuit.

The simulation results show that signals within the measurement band are transmitted to the spectrum-analyzer input with an approximately constant attenuation over frequency. At low

frequencies, the high-pass filter significantly attenuates DC and low-frequency components. At higher frequencies, the response remains stable, with no significant resonance appearing within the measurement band.

Although the RF front-end introduces a certain amount of attenuation, this is an intentional trade-off to improve protection of the spectrum-analyzer input against overvoltage transients, while also improving impedance matching and frequency-response stability. Consequently, conducted-noise signals can be delivered to the measurement instrument with a stable response, while the risk of damaging its RF input during testing is reduced.

3.5 PCB Layout

After the schematic design was completed and verified through simulation, the circuit was implemented on a two-layer FR-4 PCB. Components were arranged according to the signal-flow sequence to minimize interconnection lengths and reduce the influence of parasitic elements.

The top layer is used primarily for component placement and signal routing. The power path is designed with sufficiently wide traces to carry the required current, while the RF front-end is placed close to the BNC connector to minimize losses along the RF path.

The bottom layer is almost entirely covered with copper and serves as a continuous ground plane. This structure provides a low-impedance return path, helps reduce electromagnetic interference, and improves the transmission characteristics of RF signals.

Figures 3.9 and 3.10 show the top- and bottom-layer layouts of the PCB, respectively.

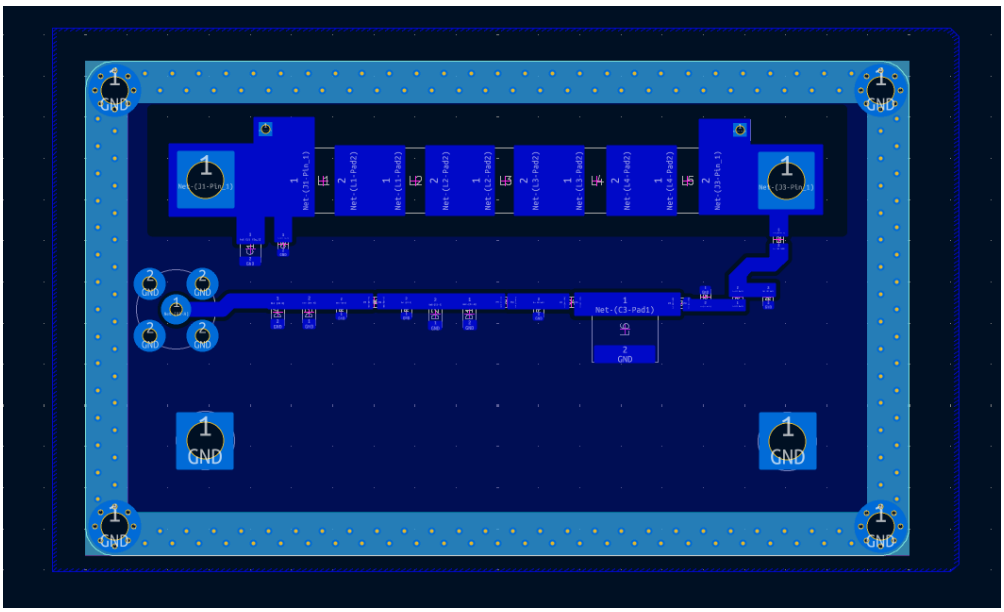


Figure 3.9: PCB top-layer layout.

A copper keepout is provided directly beneath the inductor to reduce parasitic capacitance. The practical effect of placing the inductor above a copper-filled region is discussed in the Appendix.

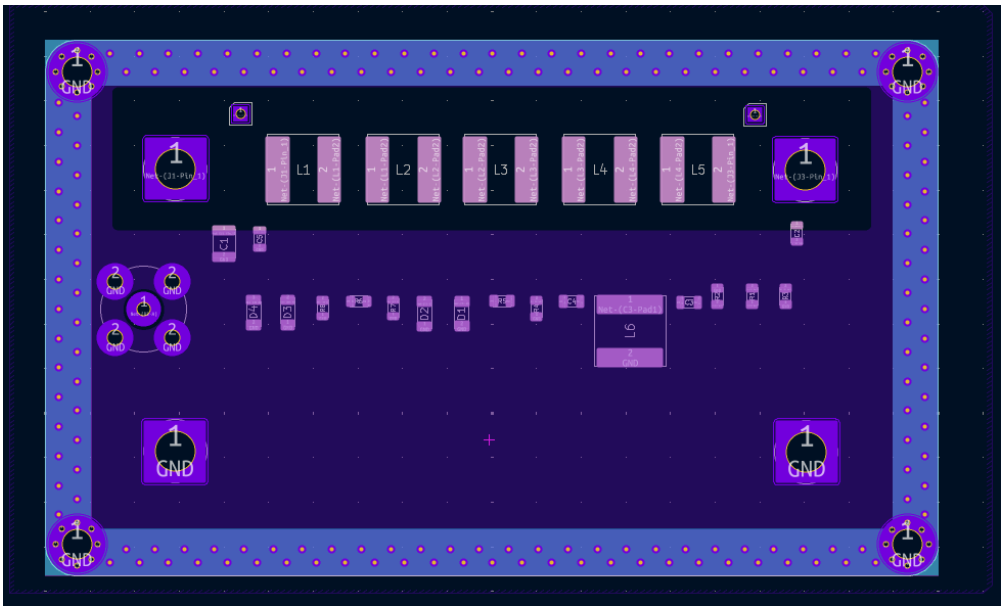


Figure 3.10: PCB bottom-layer layout.

3.6 Component Selection and Bill of Materials

After completing the design and simulation stages, the next step is to select suitable components for implementing the LISN. Although the schematic defines the circuit functionality, the actual performance of the LISN also depends on component characteristics such as parasitic inductance, parasitic capacitance, equivalent series resistance (ESR), self-resonant frequency (SRF), and current-handling capability.

For equipment operating over a frequency range from $150\,\mathrm{kHz}$ to $108\,\mathrm{MHz}$, component selection must consider not only nominal values but also high-frequency behavior, temperature stability, component availability, and manufacturing cost.

The components in this project are selected according to three main criteria:

- Compliance with the electrical requirements relevant to CISPR 25.
- Availability from suppliers in Vietnam or international distributors.
- Suitability for a prototype intended for laboratory research and testing.

The following sections describe the selection of each component group and the bill of materials (BOM) used in the design.

3.6.1 Inductor

The investigation, analysis, and selection of the inductor are discussed in detail in Section 3.2.2.

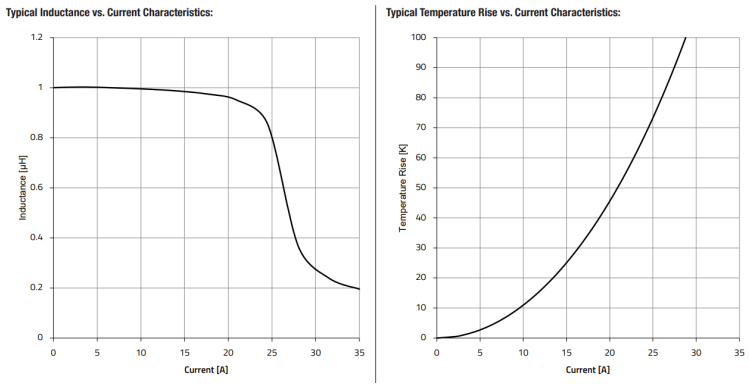


Figure 3.11: Inductance-versus-current and temperature-rise-versus-current characteristics of the Würth Elektronik WE-HCC 7443340100 inductor (Source: Datasheet 7443340100 [1]).

According to the manufacturer's datasheet, the inductor exhibits a relatively stable *inductance-versus-current* characteristic over its operating range. The inductance remains nearly constant from 0 to approximately $20\,\mathrm{A}$. At the expected system current, which is below $10\,\mathrm{A}$, the inductance remains close to its nominal value, indicating that the magnetic core is not operating near saturation. This helps maintain the impedance characteristics of the LISN and supports measurement repeatability during operation [1].

In addition, the inductor has a self-resonant frequency (SRF) of $147\,\mathrm{MHz}$, which is higher than the upper frequency limit considered for the CISPR 25 measurement range in this design ($108\,\mathrm{MHz}$). Therefore, throughout the frequency range of interest, the inductor is expected to retain its intended inductive behavior, with parasitic capacitance having a limited effect [1].

Furthermore, its maximum DC resistance (DCR) is only $3.9\,\mathrm{m\Omega}$, its rated operating current is $17\,\mathrm{A}$, and its saturation current is approximately $24\,\mathrm{A}$. These specifications meet the requirements of the LISN prototype developed in this project [1].

3.6.2 Capacitors

The capacitors in the RF signal path are selected based on capacitance, rated voltage, ESR, equivalent series inductance (ESL), self-resonant frequency (SRF), and component availability. Compared with capacitance tolerance, parasitic parameters and high-frequency behavior have a greater influence on the response of the LISN. Therefore, a tolerance of $\pm 5\%$ is considered adequate for this prototype.

Although C0G/NP0 dielectric capacitors offer better stability and lower dielectric losses, MLCCs with C0G dielectric and a capacitance of $100\,\mathrm{nF}$ are relatively limited in availability, package options, and cost. Following a component survey, a Kyocera AVX X7R MLCC (part number KAF21BR71H104JU) was selected for the RF coupling-capacitor position. The component has a capacitance of $100\,\mathrm{nF}$, a tolerance of $\pm 5\%$, an 0805 package, and a rated voltage of $100\,\mathrm{V}$. These specifications are suitable for the prototype, and the component is readily available for procurement and replacement during fabrication [5].

Since conducted-noise signals at the measurement port have very low power, the capacitor's power-handling capability is not a particularly demanding requirement. Instead, placing the

capacitor close to the RF connector, minimizing trace length, and maintaining a continuous ground plane are more important for reducing parasitic inductance and preserving the transmission characteristics of the system.

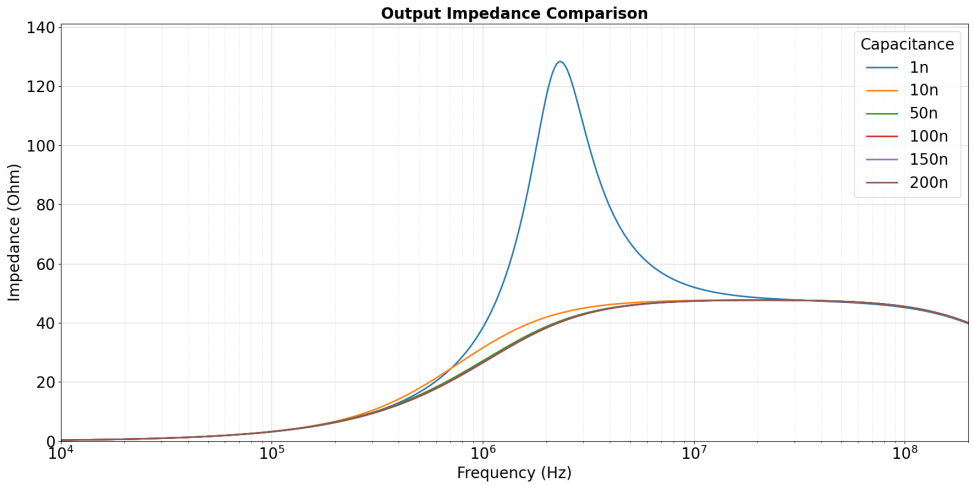


Figure 3.12: Effect of the RF coupling-capacitor value on the LISN output impedance. Simulations were performed with capacitance values ranging from $1\,\mathrm{nF}$ to $200\,\mathrm{nF}$. The results show that the impedance response changes very little over the frequency range of interest, indicating that the design is relatively insensitive to variations in the coupling-capacitor value.

Figure 3.12 shows that changing the RF coupling capacitance from $1\,\mathrm{nF}$ to $200\,\mathrm{nF}$ produces only a very small variation in the LISN output-impedance characteristic. This indicates that the circuit response is primarily determined by the $5\,\mu\mathrm{H}$ inductor and the standard $50\,\Omega$ impedance network, whereas the coupling capacitor mainly transfers noise signals to the measurement port and blocks DC components.

The simulation results also indicate that the design is relatively robust against component-value tolerance. Therefore, the selection of a $100\,\mathrm{nF}$ capacitor is not driven by a stringent capacitance requirement, but mainly by component availability, rated voltage, high-frequency characteristics, and PCB-layout considerations.

3.7 Component Selection for the Protection and Filtering Circuits

Unlike the inductor and RF coupling capacitor, the components in the attenuator, filter, and voltage-limiter circuits do not directly establish the LISN's standardized impedance. Their primary function is to condition the RF signal before it reaches the spectrum analyzer and to protect the measurement instrument from abnormal voltage transients or signals exceeding its allowable input level.

Since these functional blocks do not directly determine the LISN impedance characteristic, component selection is mainly based on accuracy, high-frequency performance, and reliability rather than strict nominal-value requirements.

3.7.1 Resistors

Resistors in the attenuator and filter circuits are used to establish the required attenuation and provide impedance matching between circuit stages. To ensure that the attenuation ratio is close to the designed value, thin-film resistors with a tolerance of $1\,\%$ are selected for the attenuator and filter circuits.

Because the RF signal power at the measurement port is very low, high resistor power ratings are not required. Resistors rated at $0.1\,\mathrm{W}$ or $0.125\,\mathrm{W}$ are sufficient for this application. Standard SMD resistors with values such as $16\,\Omega$ and $300\,\Omega$, as specified by the design, are readily available from electronics suppliers.

3.7.2 Capacitors

The capacitors in the filter determine its cutoff frequency and attenuate signal components outside the desired frequency band. Since these capacitors do not directly participate in the LISN's standardized impedance network, capacitance tolerance causes only a small shift in the filter cutoff frequency and does not significantly affect the overall system response.

The design uses multilayer ceramic capacitors (MLCCs) with X7R dielectric and a tolerance of $5\,\%$. These capacitors have low ESR and ESL, compact dimensions, and are easy to place on the PCB, making them suitable for the operating frequency range of the protection circuit.

In addition to electrical characteristics, component availability is an important design consideration. The selected X7R MLCCs are standard components supplied by many manufacturers and are readily available from electronic-component distributors. This reduces manufacturing cost, shortens procurement time, and simplifies future reproduction and maintenance of the device.

3.7.3 Voltage-Clamping Diodes

The limiter circuit protects the spectrum-analyzer input from excessive voltage transients that may occur during testing or as a result of a DUT fault. Under normal operating conditions, the RF signal voltage at the measurement port has a very small amplitude, so the diodes remain reverse-biased and have almost no effect on the signal path. The limiter begins to conduct only when the voltage exceeds the diode conduction threshold, thereby limiting the voltage applied to the measurement instrument.

In this design, the 1N4148 high-speed switching diode is selected for the limiter circuit. It is inexpensive, widely available, and has a switching time on the order of a few nanoseconds, making it suitable for the intended protection function. Although Schottky diodes or dedicated RF limiter diodes may offer lower junction capacitance and better high-frequency characteristics, the 1N4148 is considered adequate for the current design revision. Dedicated RF diodes may be evaluated in future revisions to further reduce the influence of the limiter on the measured signal.

Chapter 4 LISN Performance Verification

After fabrication, the LISN is subjected to a series of measurements to evaluate how closely the physical circuit meets the design objectives and simulated behavior described in Chapter 3. The tests focus on important LISN characteristics, including output impedance, transmission loss, and isolation between the power source and the Device Under Test (DUT).

The measured results are compared with the corresponding simulation results to evaluate the effects of component tolerances, parasitic parameters, and PCB fabrication variations on circuit performance.

The following parameters are evaluated:

1. Output impedance;
2. Insertion loss;
3. Isolation between the source and the DUT (input-to-output isolation).

For the impedance measurement, in addition to comparison with the simulation results, the measured impedance is evaluated against the impedance requirements applicable to a LISN used for CISPR 25 measurements. Specifically, the test is performed over the frequency range from $150\,\mathrm{kHz}$ to $108\,\mathrm{MHz}$, with a nominal impedance of $50\,\Omega$ at the measurement port. The measured impedance magnitude must remain within the permitted limits of the reference characteristic throughout this frequency range.

The measurement methods, instrument connection diagrams, and test conditions for each measurement are described in the following sections.

4.1 Measurement-System Setup

A measurement system is configured to evaluate the characteristics of the fabricated LISN under RF-port connection conditions representative of the intended measurement setup. The system connection diagram is shown in Figure 4.1.

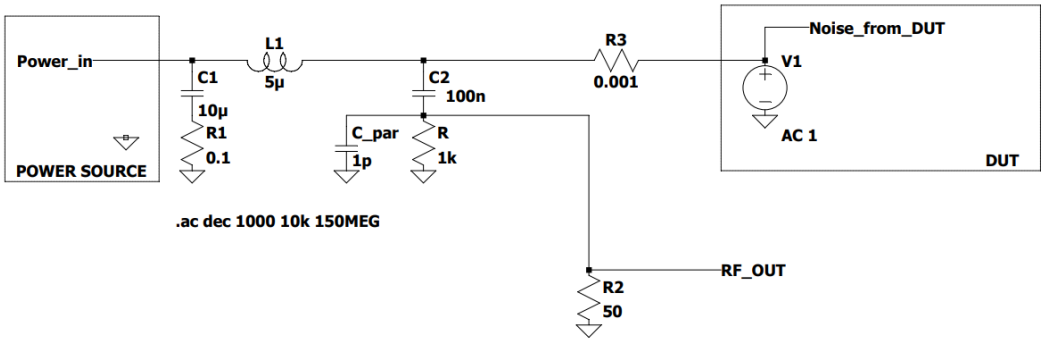


Figure 4.1: LISN measurement-system setup.

The measurement system uses an RF network analyzer, coaxial cables, and standard $50\,\Omega$ terminations. Before each measurement, the system is calibrated to reduce the influence of measurement cables and auxiliary components on the results.

A photograph of the completed measurement setup can be included in this section to illustrate the arrangement of the LISN, measurement instrument, and RF connections.

4.1.1 Output-Impedance Measurement

After fabrication, the LISN must be characterized to verify the impedance characteristics of the physical circuit. This is an important step in evaluating the agreement between the design, the simulation results, and the assembled hardware.

For the $5\,\mu\mathrm{H}$ LISN configuration used for conducted-emission measurements according to CISPR 25, the impedance seen from the DUT port is expected to remain close to the nominal value of $50\,\Omega$ over the frequency range from $150\,\mathrm{kHz}$ to $108\,\mathrm{MHz}$. In practice, the measured impedance may deviate from the model because of component tolerances, parasitic inductance and capacitance, PCB trace geometry, connectors, and cables.

A Vector Network Analyzer (VNA) is used to characterize the impedance of the LISN. Compared with a method that uses a separate excitation source and independent voltage and current measurements, a VNA one-port reflection measurement provides a more direct way to acquire the reflection coefficient and convert it into impedance, simplifying data collection.

Measurement Setup

The measurement is performed using a one-port reflection method. The VNA port is connected directly to the DUT port of the LISN through a coaxial cable. The center conductor of the cable is connected to the signal terminal at the DUT port, while the cable shield is connected to the corresponding GND terminal of the circuit.

Before measurement, the VNA must be calibrated for a one-port measurement at the connection plane of the LISN. Calibration compensates for the measurement cable and systematic errors to the extent supported by the calibration standards. After calibration, the measurement frequency range is set from $150\,\mathrm{kHz}$ to $108\,\mathrm{MHz}$, corresponding to the frequency range used to evaluate the LISN impedance.

During the measurement, the remaining LISN ports must be maintained under termination conditions consistent with the design and simulation configuration. In particular, the RF port must be terminated with a standard $50\,\Omega$ load, while the power-supply port is maintained in the connection configuration used in the simulation model. Consistent termination conditions are necessary because the impedance observed at the DUT port depends on the loads connected to the other ports.

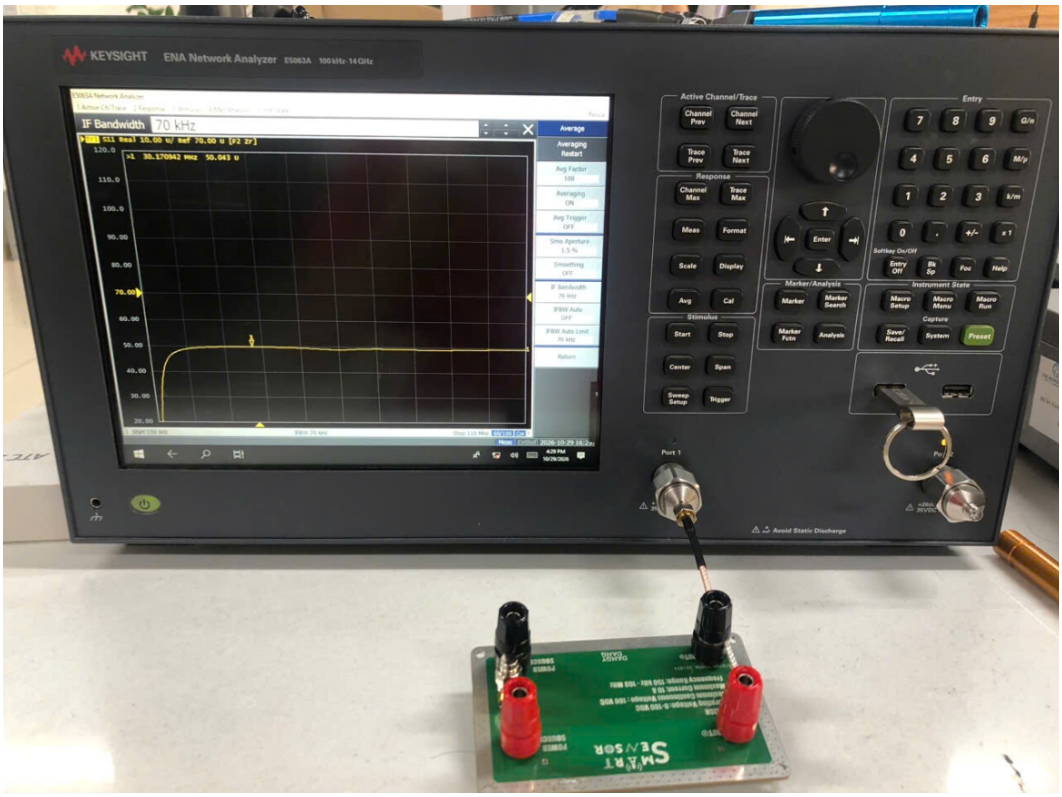


Figure 4.2: VNA connection for measuring the impedance seen from the DUT side of the LISN.

The VNA applies a low-power RF signal to the DUT port and measures the reflected signal. From this measurement, the instrument determines the complex reflection coefficient S_{11} at each frequency point. The impedance at the measurement plane is calculated using:

$$Z_{\text{out}}(f) = Z_0 \frac{1 + S_{11}(f)}{1 - S_{11}(f)} \tag{4.1}$$

where:

- $Z_{\text{out}}(f)$ is the impedance seen from the DUT port;
- $Z_0 = 50\,\Omega$ is the characteristic impedance of the measurement system;
- $S_{11}(f)$ is the complex reflection coefficient measured at the VNA port.

From the complex impedance, the magnitude is calculated as:

$$|Z_{\text{out}}(f)| = \sqrt{\text{Re}^2\{Z_{\text{out}}(f)\} + \text{Im}^2\{Z_{\text{out}}(f)\}} \tag{4.2}$$

Thus, a single VNA port provides the impedance characteristic as a function of frequency without requiring a separate excitation source and current-measurement circuit, as in the AC simulation method.

Note that this measurement is performed at the DUT port. It is therefore essential to ensure that no DC voltage or high-power signal reaches the VNA RF port. If the LISN is energized during measurement, an appropriate isolation or coupling arrangement must be used to protect the instrument.

Evaluation Criteria

The measurement results are evaluated based on two main considerations.

First, the measured impedance characteristic is compared with the LTspice simulation results presented in Chapter 3. This comparison is intended to evaluate the agreement between the circuit model and the physical hardware and to identify frequency regions with significant deviations.

Second, the impedance characteristic is compared with the reference limit curve used to evaluate the LISN according to CISPR 25. The comparison is performed over the frequency range from $150\,\text{kHz}$ to $108\,\text{MHz}$, with emphasis on the impedance magnitude seen from the DUT port.

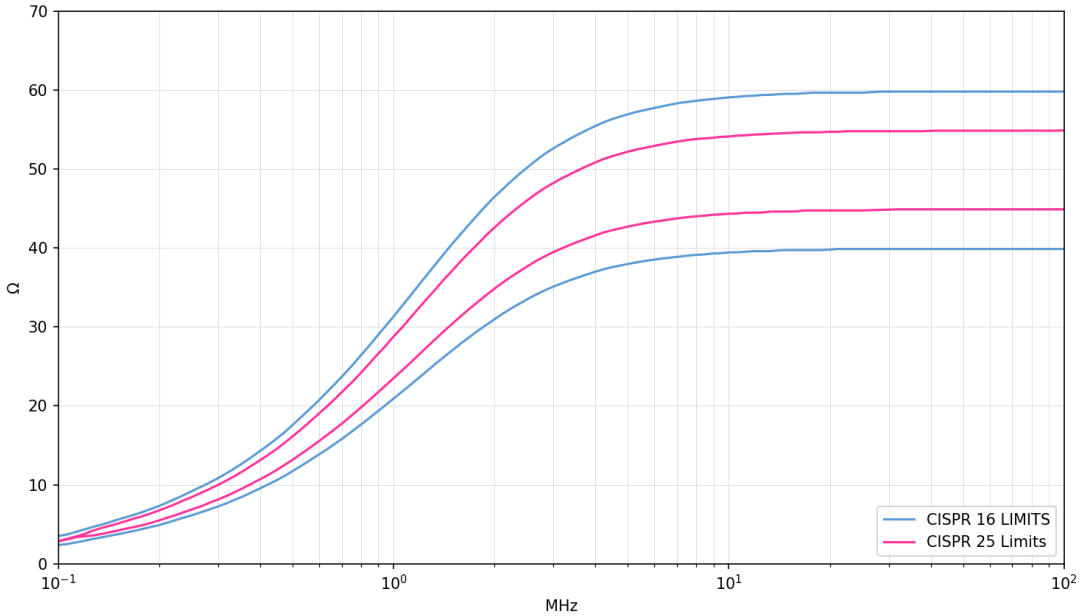


Figure 4.3: Reference impedance-limit curves used for LISN evaluation.

For presentation, the simulated characteristic, measured characteristic, and reference limit curves are plotted on the same graph. This makes it possible to directly observe deviations between the model and the hardware and to determine whether the impedance remains within the required limits over the frequency range under investigation.

Measurement Results

The measured impedance of the fabricated LISN is shown in Figure 4.4. The VNA trace shows the impedance variation with frequency at the DUT port.

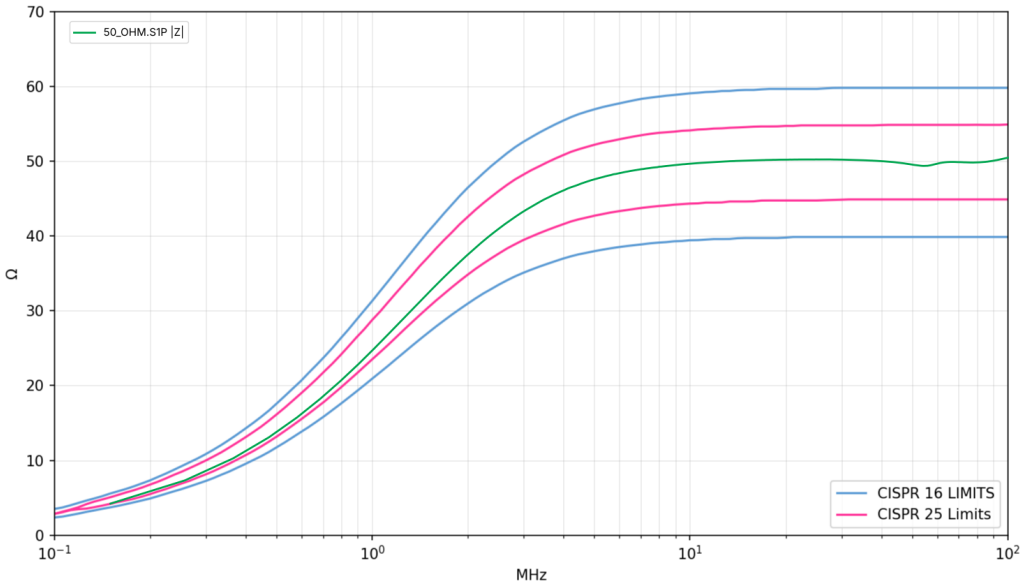


Figure 4.4: Impedance characteristic seen from the DUT side of the LISN, measured using a VNA.

The measurement results show that the device remains within the specified limit region and satisfies the $50\,\Omega$ impedance test.

4.1.2 RF Filter Characterization

The RF filter is used at the LISN measurement output to attenuate unwanted signal components and protect the measurement instrument from signals with excessive amplitude or frequencies outside the intended measurement band. Since the filter response depends on frequency and the actual characteristics of its components, post-fabrication characterization is necessary to verify its performance.

The filter is characterized by measuring its transmission coefficient, S_{21}, using a Vector Network Analyzer (VNA). This measurement determines the signal attenuation through the filter as a function of frequency, allowing its passband transmission and attenuation outside the desired frequency range to be evaluated.

Measurement Setup

To characterize the transmission response, the filter is connected between the two RF ports of the VNA. Port 1 provides the excitation signal, while Port 2 receives the signal transmitted through the filter. Both measurement ports have a characteristic impedance of $50\,\Omega$, consistent with the impedance-matched conditions of the RF measurement system.

Before measurement, the VNA is calibrated using a two-port calibration at the connection planes of the filter. This calibration reduces the influence of measurement cables, connectors, and systematic errors on the results. After calibration, the filter is connected directly between the two measurement ports in the same signal direction intended for its integration into the LISN.

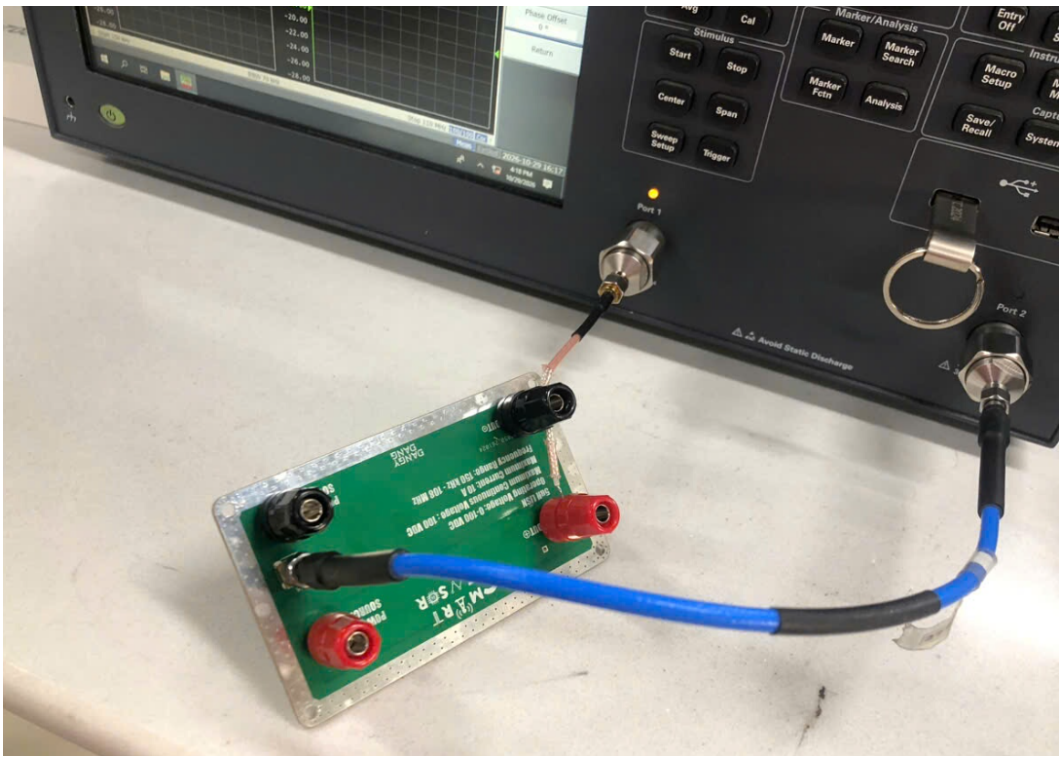


Figure 4.5: VNA connection diagram for measuring the RF filter transmission characteristic.

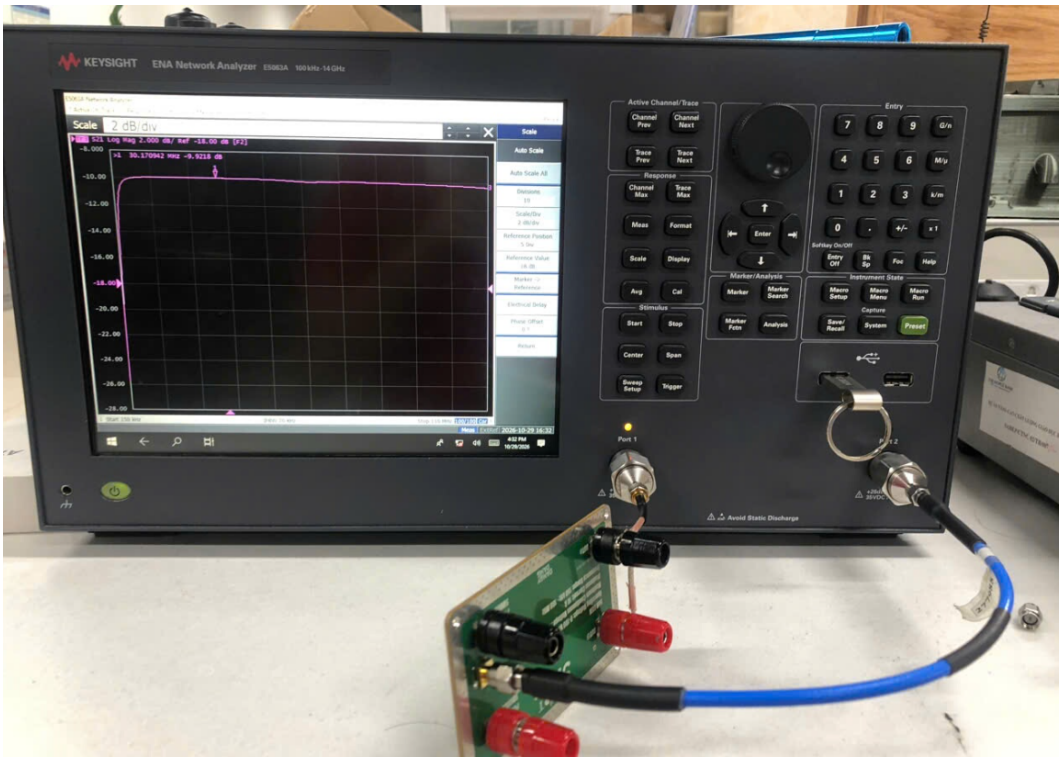


Figure 4.6: VNA connection diagram for measuring the RF filter transmission characteristic.

During measurement, the VNA transmits an RF signal from Port 1 and measures the signal received at Port 2. The ratio between the transmitted wave at Port 2 and the incident wave at Port 1 is represented by the transmission coefficient S_{21}.

The transmission characteristic is commonly expressed in decibels as:

$$S_{21,\mathrm{dB}}(f) = 20 \log_{10} |S_{21}(f)| \tag{4.3}$$

The frequency sweep is configured to match the operating range of the system. In particular, the range from $150\,\mathrm{kHz}$ to $108\,\mathrm{MHz}$ is used to evaluate the filter within the conducted-emission measurement band specified for CISPR 25.

Measurement Results

The measured transmission characteristic of the filter is presented in Figure 4.7. The trace shows the variation of the transmission coefficient with frequency, indicating the frequency regions in which signals are passed or attenuated by the filter network.

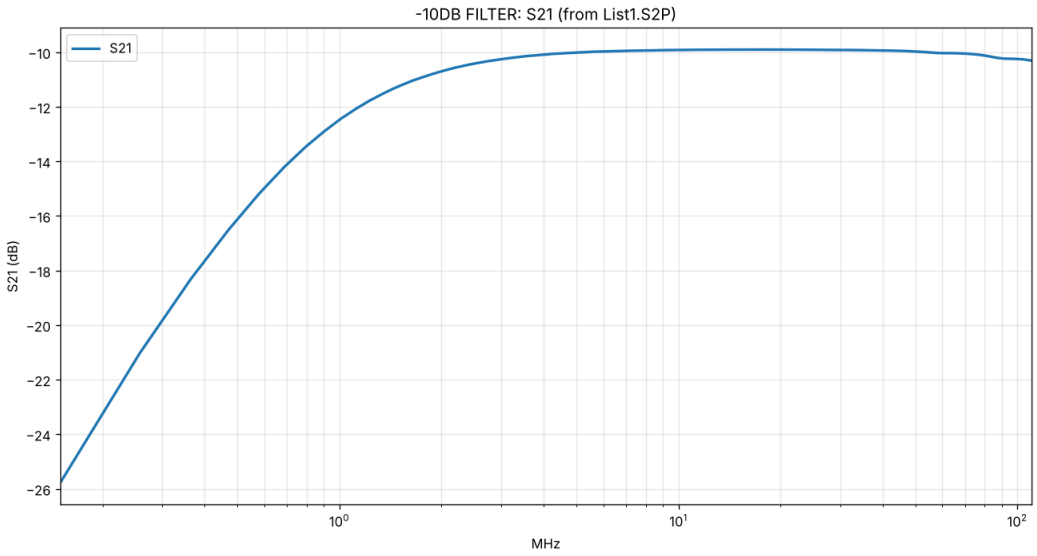


Figure 4.7: RF filter transmission characteristic (S_{21}) measured using a VNA.

The measurement shows that the RF output signal is attenuated by approximately $10\,\mathrm{dB}$, as intended by the design and simulation.

4.1.3 Isolation Measurement Between the Power Source and the DUT

One of the important functions of a LISN is to limit the transmission of high-frequency noise from the equipment under test (DUT) back into the power-supply system. Without sufficient isolation, noise generated by the DUT may propagate along the supply line, affect the measurement environment, and distort the conducted-emission results.

Therefore, in addition to the impedance characteristic at the DUT port, the isolation between the source port and the DUT port must also be evaluated after fabrication. This measurement determines the attenuation experienced by a signal transmitted from the DUT side toward the power-supply side over the frequency range of interest.

Measurement Setup

Isolation is characterized using a Vector Network Analyzer (VNA) through a two-port transmission measurement. In this configuration, VNA Port 1 is connected to the DUT side, while Port 2 is connected to the power-supply side of the LISN.

Before measurement, the VNA is calibrated using a two-port calibration at the connection planes of the device. The remaining LISN ports are maintained under termination conditions appropriate to the measurement configuration. In particular, the RF port of the LISN must be terminated with a standard $50\,\Omega$ load to represent the operating condition of the noise-measurement instrument.

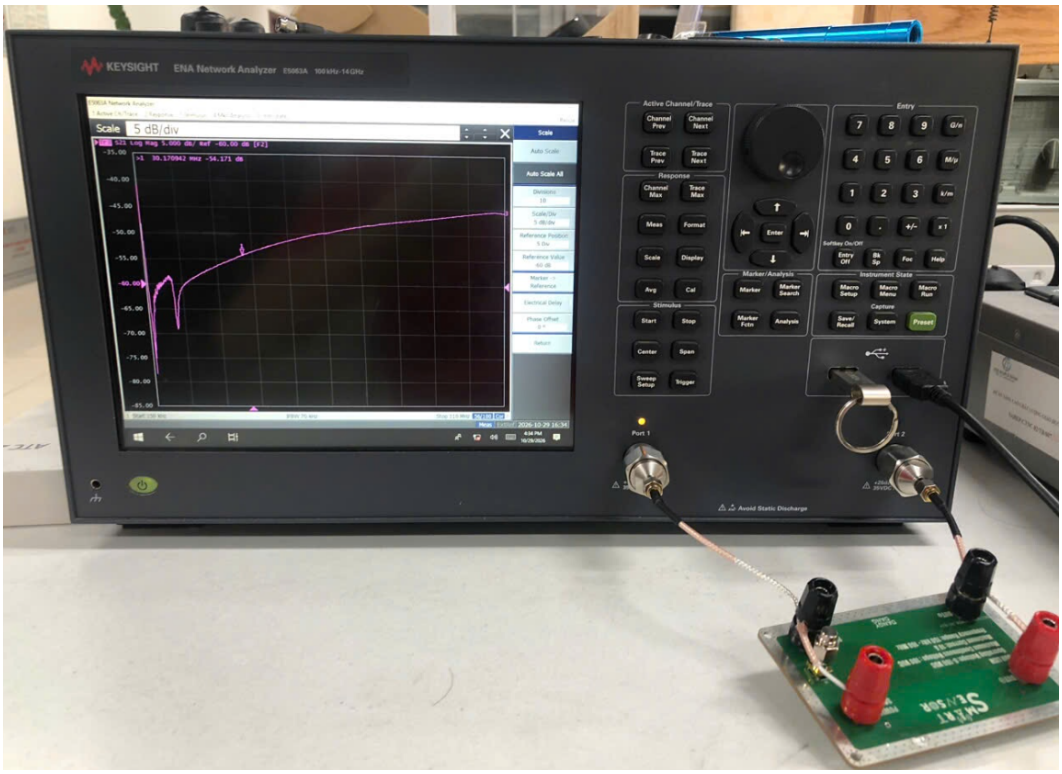


Figure 4.8: VNA connection diagram for evaluating isolation between the DUT port and the LISN source port.

During measurement, the VNA transmits a signal from Port 1 and records the signal received at Port 2. The transmission coefficient between the two ports is represented by S_{21}:

$$S_{21}(f) = \frac{b_2(f)}{a_1(f)} \tag{4.4}$$

where:

- $a_1(f)$ is the incident wave at Port 1;
- $b_2(f)$ is the transmitted wave at Port 2;
- $S_{21}(f)$ is the transmission coefficient from the DUT side toward the power-source side.

Measurement Results

The measured isolation between the DUT port and the source port is shown in Figure 4.9. The trace represents transmission attenuation as a function of frequency and indicates the ability of the LISN to limit noise propagation back toward the power source.

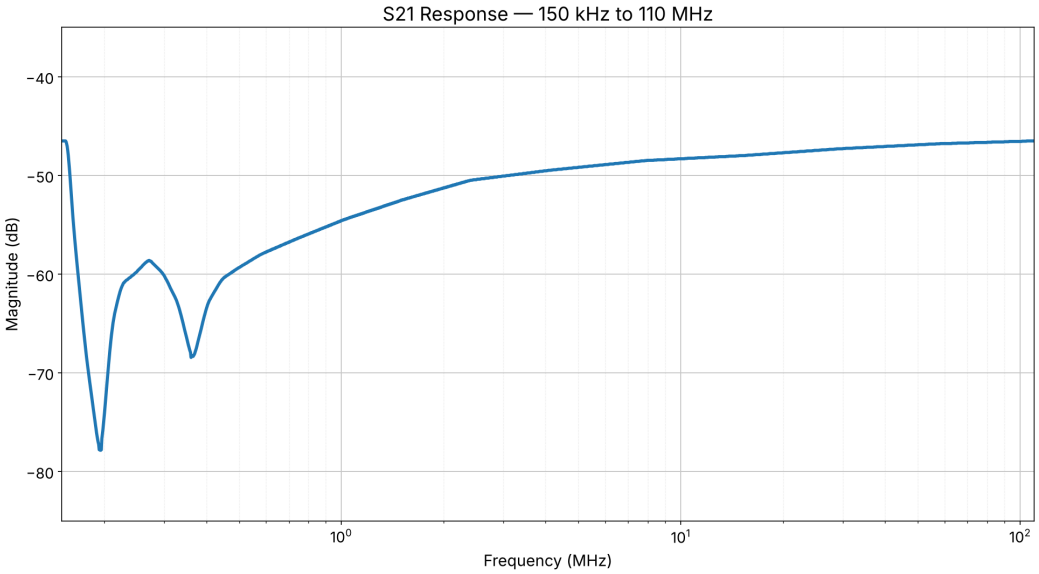


Figure 4.9: Isolation characteristic between the DUT port and the LISN source port, measured using a VNA.

Over the frequency range of interest, the signal transmitted from the DUT side to the source side remains below $-45\,\mathrm{dB}$. This indicates a high level of attenuation under the measurement configuration and suggests that the LISN is suitable for EMI/EMC pre-compliance measurements.

Chapter 5 Appendix

5.1 Effect of Parasitic Capacitance in the PCB Layout

During the fabrication and evaluation of the LISN, a practical issue was identified in the initial PCB layout: parasitic capacitance between the inductor and the GND copper plane beneath it. This effect is not fully represented in the schematic, but it can significantly influence circuit behavior at higher frequencies.

5.1.1 Cause of the Problem

In the initial PCB design, the $5\,\mu H$ inductor was placed above a relatively continuous GND copper region on the layer below. Because the spacing between the inductor and the copper plane was small, a distributed parasitic capacitance was formed between them.

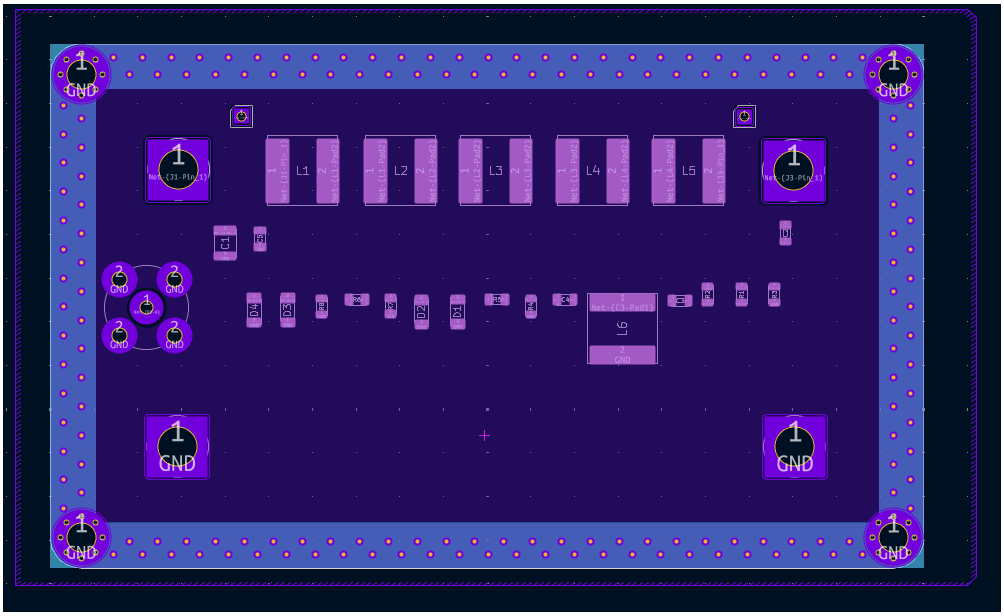
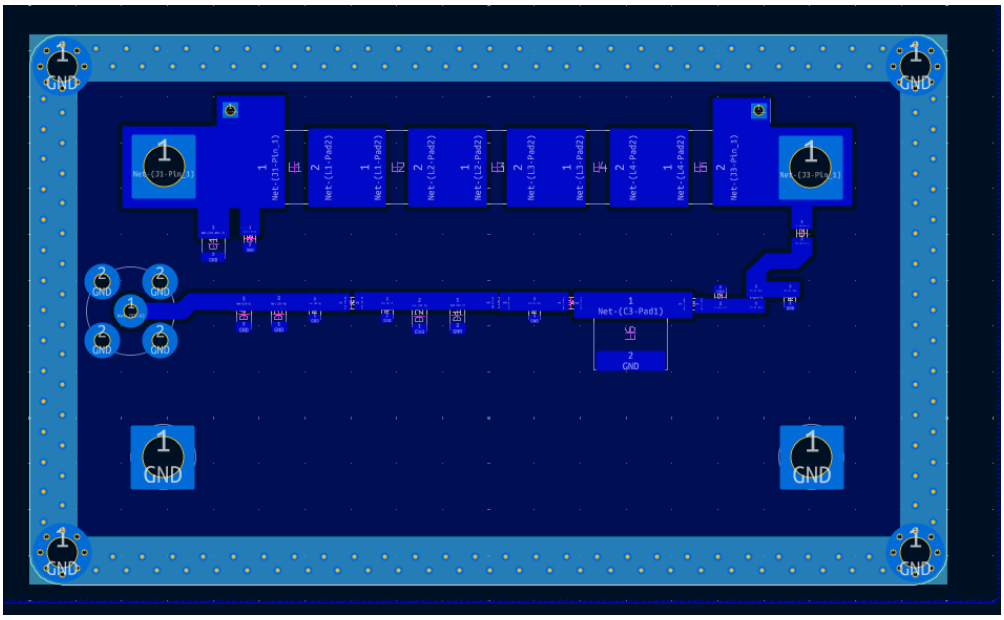


Figure 5.1: Initial PCB top-layer layout.

From a physical perspective, this structure can be approximated as a capacitor, with the inductor acting as one electrode and the GND plane acting as the other. The capacitance depends on the effective overlapping copper area, the separation between PCB layers, and the dielectric properties of the insulating material.


GND
GND
GND

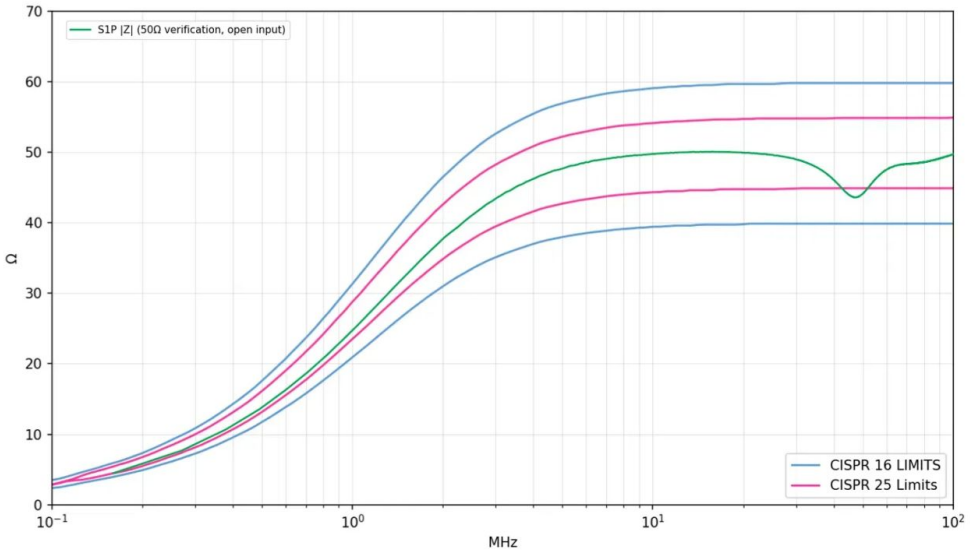


Figure 5.4: Effect of parasitic capacitance on the measured $50\,\Omega$ impedance characteristic.

5.1.2 Corrective Action

To reduce the influence of parasitic capacitance, the GND copper directly beneath the inductor was removed by scraping away the copper in the corresponding PCB area. After this modification, the overlapping copper area was reduced, thereby decreasing the capacitive coupling between the inductor and the GND plane.



Figure 5.5: GND copper removed from the area directly beneath the inductor to reduce parasitic capacitance.

This modification was carried out directly on the fabricated hardware without changing the nominal value of the inductor or any other circuit component. Removing the copper region beneath the inductor reduced capacitive coupling to the GND plane and was intended to limit the influence of this parasitic element on the resonant behavior of the network.

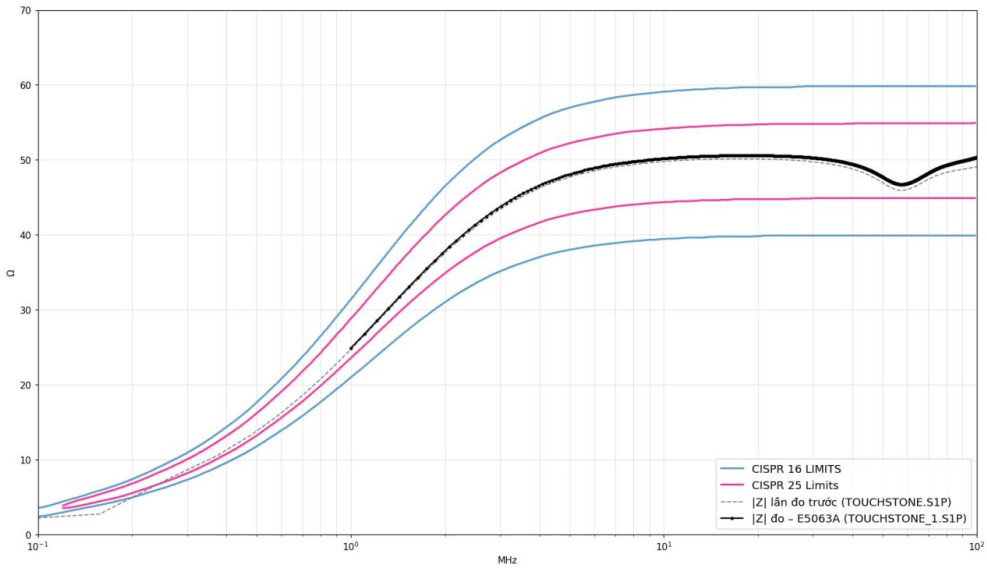


Figure 5.6: Measurement result after removing the copper beneath the inductor.

5.1.3 Design Lessons

This issue highlights several important considerations when placing inductive components in RF circuits and conducted-emission measurement equipment:

- The nominal inductance alone should not be considered; parasitic capacitance introduced by the PCB structure must also be taken into account.
- Pay attention to the GND plane and copper regions directly beneath an inductor, particularly when the inductor is used in a network with high-frequency performance requirements.
- A copper keepout beneath the inductor should be considered during the PCB design stage when permitted by the component structure and electrical requirements.
- After modifying the PCB layout, the impedance characteristics and frequency response should be measured again to verify the effectiveness of the modification.

This experience demonstrates that parasitic effects introduced by PCB layout can directly affect LISN performance, even when the schematic and nominal component values remain unchanged. Therefore, considering the physical structure of the circuit and validating it through practical measurements are important parts of RF hardware design.

References

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